



DATA ANALYTICS PROJECT

TerraTrac Insights

Demand Forecasting & Supply Planning — Case Study

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GitHub: [@tahtabirrahman](#)

Repository: github.com/tahtabirrahman/terratrac-insights

Live dashboard: trac-planner-insight.lovable.app

25 SKUs · 4 distribution centers · 5 suppliers · 104 weeks of history · 220 purchase orders

What This Project Is

A self-directed data analytics case study: I took a synthetic supply-chain dataset, scoped the analytical problem, built a working demand-forecasting and inventory-planning product from it, and shipped it as a live, published dashboard.

Business context

TerraTrac Equipment Parts is a fictional distributor of spare parts for heavy construction equipment (hydraulics, engine & powertrain, undercarriage, electrical/cabin, ground-engaging tools). It operates four regional distribution centers and sources from five category-specialist suppliers with different lead times and reliability. All company names and figures in the dataset are synthetic, built for this exercise.

Problem statement

TerraTrac's planning process had no formal reorder-point discipline. Nobody could answer, with evidence: which SKU/warehouse positions are actually at stockout risk right now, which suppliers' reliability claims hold up against real delivery data, and whether current demand forecasts are usable or just noise.

Project objective

Build a 12-week demand forecast, a safety-stock and reorder-point plan, and a supplier-risk assessment — grounded in the actual purchase-order history rather than stated supplier claims — and present it as a tool a supply chain leadership team could actually use.

Key questions the analysis answers

- Which SKU × warehouse positions are below their reorder point today, and how urgently?
- Which supplier carries the most lead-time risk, and does the data support the supplier master's own reliability claims?
- How accurate is a trend + seasonality + promotion forecast, measured honestly on a blind holdout?
- What should be reordered this cycle, respecting MOQ and pack size?

Expected outcome

A working, interactive dashboard covering forecast accuracy, inventory risk flags, a replenishment plan, and a supplier scorecard — published and publicly reviewable on GitHub.

Why This Problem Matters

Heavy-equipment parts distribution runs on thin margins for error: a stocked-out hydraulic seal or undercarriage part can idle a customer's construction equipment, and lead times of 2-5 weeks mean a planning mistake isn't quickly correctable. Two failure modes are common in under-instrumented supply operations, and both show up in this dataset:

Silent under-stocking

Inventory drifts below what's needed to cover a lead time, and nobody notices until a stockout actually happens, because there's no reorder-point discipline flagging it in advance.

Unverified supplier claims

Reliability figures used for planning come from supplier scorecards that are rarely checked against what the purchase-order data actually shows.

Objectives

- 1. Produce a defensible 12-week demand forecast per SKU, correctly handling known data issues (promotions, a stockout-censored demand period) rather than ignoring them.
- 2. Convert that forecast into a safety-stock and reorder-point plan with a stated service-level assumption.
- 3. Flag stockout-risk and overstock positions, and produce a replenishment recommendation that respects MOQ and pack size.
- 4. Score supplier lead-time and reliability risk using actual purchase-order history, not just the stated figures.

Business questions this analysis answers

Which positions need action this week?	Supply Planning tab — reorder point, safety stock, and a risk flag per SKU x warehouse.
Can we trust the forecast?	Reported honestly against a naive baseline, on an out-of-sample holdout the model never saw.
Which supplier is the real risk?	Supplier Risk tab — lead time and on-time rate computed from 220 actual purchase orders.

Expected analytical outcome

A forecast that measurably beats a naive baseline; a reorder-point plan grounded in a stated, defensible service-level assumption; and a supplier risk ranking backed by transaction-level evidence rather than a vendor's own scorecard.

The Data Behind the Analysis

A synthetic case-study dataset (six linked tables in one workbook), deliberately built with realistic messiness — not clean, ready-to-plot data.

DEMAND HISTORY ROWS
10,400
25 SKUs × 4 DCs × 104 weeks

SKUS
25
5 categories, 3 ABC classes

PURCHASE ORDERS
220
across 5 suppliers

HISTORY SPAN
104 wks
~2 years, weekly

Tables

TABLE	WHAT IT HOLDS
Product_Master	25 SKUs — category, ABC class, unit cost/price, MOQ, pack size, shelf life
Warehouse_Master	4 distribution centers — region, storage capacity
Supplier_Master	5 suppliers — stated average lead time, lead-time variability, OTIF reliability
Demand_History	10,400 rows — weekly sell-through by SKU × warehouse, with promotion and stockout flags
Inventory_Snapshot	Current on-hand and on-order quantity by SKU × warehouse
Purchase_Order_History	220 orders — promised vs. actual delivery date, used for real lead-time and reliability analysis

Data dictionary — Demand_History (the core table)

COLUMN	TYPE	NOTES
Week_Start_Date	Date	Weekly bucket
SKU_ID / Warehouse_ID	Text	Foreign keys to Product_Master / Warehouse_Master
Units_Sold	Integer	Weekly sell-through
Unit_Price_USD	Decimal	Price that week
Promotion_Flag	Boolean	10 promotional weeks, causing real demand spikes
Stockout_Flag	Boolean	Marks a 6-week window where observed demand is artificially suppressed by a real supply disruption — a data-quality issue that has to be corrected, not ignored (see Data Cleaning)

Known data-quality issues (by design)

- Each SKU has its own trend, seasonal cycle, and noise level — no uniform pattern to exploit.
- A 6-week supply disruption (weeks 41–46) suppresses observed Undercarriage demand at the two Asia-Pacific DCs — classic demand censored by a stockout.
- On-hand inventory is noisy relative to recent demand — some positions are overstocked, others genuinely at risk.
- Supplier lead times vary in both mean and standard deviation.

Assumptions

Service level: 98% for ABC Class A SKUs, 95% for B/C — a judgment call, not given in the data, stated explicitly rather than left implicit. Replenishment review cycle: 4 weeks.

What Was Actually Used

Listed honestly, including how each tool was actually used — not a generic tools list.

ANALYSIS

Python — pandas, NumPy, SciPy/statsmodels

Used to build the analysis pipeline: demand forecasting (trend + seasonality + promotion regression), stockout-censoring correction, safety-stock/reorder-point calculation, and supplier-risk scoring from raw purchase-order data. The initial pipeline was AI-drafted from a specification; I then verified its output against the source data by hand, found two real calculation bugs (detailed on page 9), traced each to its exact line, corrected it, and re-verified before publishing.

PRODUCT

Lovable (AI application builder)

Used to generate the interactive dashboard (React + TypeScript + Tailwind) from a written specification, wired to the analysis output as a static JSON data file. I directed the build, requested specific fixes and layout changes through it, and managed the publish/deploy cycle.

VERSION CONTROL

GitHub

Repository hosting, commit-level data fixes, file organization, and public deployment of the corrected dashboard and analysis script.

Not used in this project

SQL · Power BI · Excel-based analysis · Jupyter Notebook · Tableau — not part of this project's actual toolchain, so no fabricated section for any of them appears in this document.

Handling the Censored-Demand Problem

The dataset's central data-quality challenge: a real 6-week supply disruption suppressed observed demand for Undercarriage SKUs at the two Asia-Pacific DCs. Reported sell-through in that window understates true demand — it's censored, not soft.

The three options, and why I picked one

OPTION	PROBLEM WITH IT
Ignore it	Forecast and safety stock both come out too low for the SKUs and warehouses that need the buffer <i>most</i> .
Exclude those weeks entirely	Shrinks the training sample and biases the seasonal estimate for that calendar period.
Correct it (chosen)	Reconstructs a defensible estimate of true demand without discarding data.

Method: sibling-warehouse ratio correction

For each affected SKU, I computed the average ratio of that Asia-Pacific DC's demand to the combined demand at the two unaffected DCs (Dallas, Rotterdam), using only clean, non-promotional, non-censored weeks as the basis. That ratio was then applied to the unaffected DCs' demand during the censored window to estimate what the affected DC would have sold without the disruption.

Result: measured Undercarriage demand during the censored weeks was suppressed by an average of **163%** relative to the corrected estimate. Left uncorrected, this would have understated both the forecast and the safety-stock requirement for the highest-risk product category in the portfolio.

Other preprocessing steps

Promotion handling: 10 known promotional weeks cause genuine demand spikes. Rather than let them distort the trend/seasonal fit, I excluded them from the baseline fitting step and estimated a separate promotion-uplift multiplier from those weeks specifically.

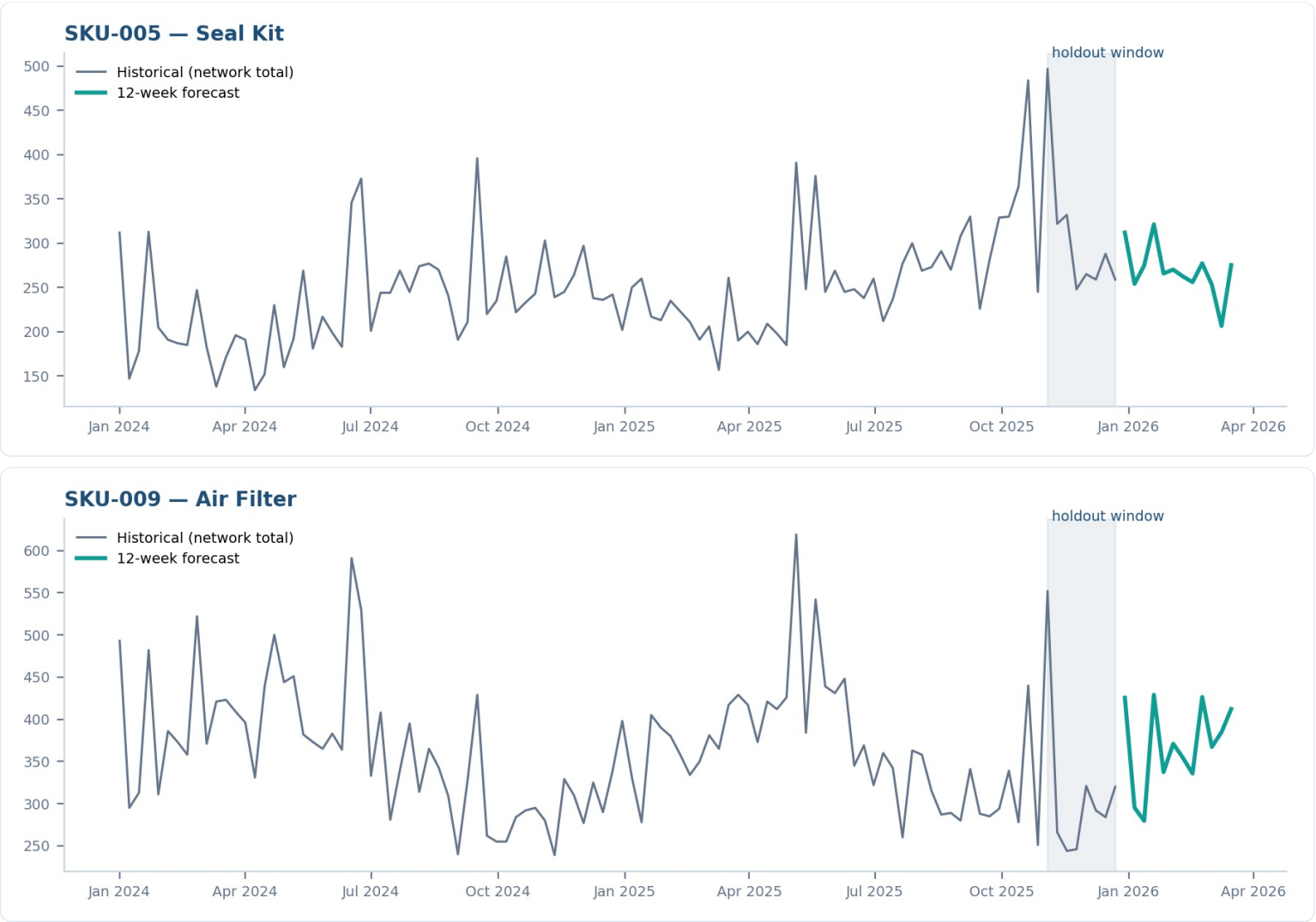
Data validation: cross-checked corrected demand totals, the stockout-flag date range, and supplier reliability figures against the raw source tables rather than trusting intermediate output at face value — this is what surfaced the two calculation bugs described on page 9.

No missing-value imputation or deduplication was required — the source tables were structurally complete; the data-quality issues here are about correctness of interpretation (censored demand, unverified supplier claims), not incomplete records.

EXPLORATORY & FORECASTING ANALYSIS

Demand Patterns and the Forecasting Approach

Each SKU carries its own trend, seasonal cycle, and noise level — there is no single pattern that fits all 25. The charts below are the network-total weekly history for three of the five SKUs in scope, with the 12-week forecast and the blind holdout window marked.



Method

Trend × seasonality × promotion-uplift regression, fit per SKU × warehouse on non-promotional weeks. Chosen over ARIMA/ETS given the data: 104 weeks is barely two full seasonal cycles, and each SKU's noise profile differs enough that a transparent, auditable model is more defensible to a planning audience than a marginally-tunable black box.

Validation

WMAPE, 8-WEEK BLIND HOLDOUT 23.1% model, not trained on this window	NAIVE BASELINE (52-WK AVG) 24.0% what it's compared against	HOLDOUT LENGTH 8 wks true out-of-sample test
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The model is reported honestly against a naive baseline on data it never trained on — not just measured on the same data it was fit to, which would overstate accuracy.

Core Calculations

Two representative pieces of the analysis pipeline — not every line, just what demonstrates the actual analytical logic.

Safety stock & reorder point

Accounts for **both** demand variability and supplier lead-time variability — not just demand variability alone, which is the more common (and less complete) version of this formula:

```
safety_stock = Z * sqrt(
    lead_time_weeks * demand_std**2
    + avg_weekly_demand**2 * lead_time_std_weeks**2
)
reorder_point = avg_weekly_demand * lead_time_weeks + safety_stock

# Z = 2.05 (~98% service) for ABC Class A, 1.65 (~95%) for B/C
# lead_time_weeks / lead_time_std come from actual PO history, not the supplier's stated average
```

Stockout-censoring correction

The sibling-warehouse ratio method described on page 6, as implemented:

```
# true censored weeks, from the Stockout_Flag column directly
censored_weeks = demand_history.loc[demand_history.Stockout_Flag == 1, 'Week_Start_Date'].unique()

sibling_avg = (demand_NA1 + demand_EU1) / 2
clean_weeks = ~(is_censored | is_promotion)
ratio = (demand_AP[clean_weeks] / sibling_avg[clean_weeks]).mean()

corrected_demand = ratio * sibling_avg[censored_weeks]
```

Why this matters more than the code itself

The formula and the correction method are both defensible on their own. What made this a real analytical exercise wasn't writing this code — it was verifying that the actual pipeline (AI-drafted from a specification) implemented them correctly. It didn't, in two places. That's the subject of the next page.

Two Bugs Found by Checking the Output Against the Data

This is the part of the project that actually demonstrates analytical judgment: not writing code, but not trusting it either — checking derived numbers against source data and catching what didn't add up.

Bug 1 — four suppliers with an "identical" 70% on-time rate

The supplier scorecard showed Trackline, Circuit & Cab, Vantage, and Ironclad all at exactly 70% on-time delivery, with Meridian at 60%. Four independently-measured suppliers landing on the exact same figure is not a real-world coincidence — it's a rounding artifact. Traced to this:

```
- sup_actual = po.groupby("Supplier_ID").agg(  
-     OnTime_Pct_Actual=("On_Time_Flag", "mean"),  
- ).round(1)                                ## rounds the 0-1 FRACTION to 1 decimal first  
- sup_actual["OnTime_Pct_Actual"] *= 100      ## then multiplies -- precision already lost
```

```
+ sup_actual = po.groupby("Supplier_ID").agg(  
+     OnTime_Pct_Actual=("On_Time_Flag", "mean"),  
+ )                                ## no premature rounding  
+ sup_actual["OnTime_Pct_Actual"] = (sup_actual["OnTime_Pct_Actual"] * 100).round(1) ## round AFTER scaling
```

Rounding a 0–1 fraction to one decimal place before multiplying by 100 collapses any value into a 10-point bucket — 0.583, 0.651, 0.660, 0.683 all round to 0.7. Fixing the order of operations recovered the real, distinct figures: **65.1%, 58.3%, 66.0%, 68.3%, 73.7%**.

Bug 2 — an off-by-one week range in the stockout correction

The censored-week correction (page 6) used a hardcoded date range instead of reading it from the data's own Stockout_Flag column:

```
- CENSORED_WEEKS = range(40, 46) ## weeks 40-45 -- one week early
```

```
+ CENSORED_WEEKS = range(41, 47) ## weeks 41-46 -- matches Stockout_Flag == 1
```

This silently corrected one week that was never actually censored, and missed the real sixth censored week entirely — wrong on both ends of the range.

What this demonstrates

Both bugs were caught the same way: by treating a generated number as a claim to verify, not a fact to accept — cross-checking supplier figures against each other (four identical values is implausible) and checking a date range against the column that actually defines it, rather than a hardcoded assumption.

TerraTrac Equipment Parts

Demand Forecasting & Supply Planning

Planning cycle - week of 29 Dec 2025 - service level 95%

Overview

Demand Forecasting

Supply Planning

Supplier Risk

Executive Summary

CASE STUDY

TerraTrac Equipment Parts – Demand Forecasting & Supply Planning

TerraTrac distributes spare parts for heavy construction equipment across five categories, operating four regional distribution centers and sourcing from five category-specialist suppliers. This dashboard reviews two years of weekly sell-through, the current inventory position, and 220 historical purchase orders to produce a 12-week forecast and a replenishment plan.

FORECAST ACCURACY (WMAPE)

23.1%

vs 24% naive baseline

POSITIONS BELOW REORDER POINT

26 / 28

16 cannot cover one lead time

RECOMMENDED BUY THIS CYCLE

\$8.16M

MOQ and pack size respected

WORST SUPPLIER LEAD TIME

33.4 d

Trackline Mfg - o 10.0 d - P95 51 d

Network

Four regional distribution centers

Dallas DC	180,000 units cap.
North America - USA	
Rotterdam DC	150,000 units cap.
Europe - Netherlands	
Chennai DC	200,000 units cap.
Asia Pacific - India	
Jakarta DC	90,000 units cap.
Asia Pacific - Indonesia	

Suppliers

Category specialists, measured on actual PO history

SUPPLIER	CATEGORY	LEAD TIME	Σ	OTIF
Trackline Manufacturing	Undercarriage	33.4 d	10 d	65.1%
Meridian Hydraulics Ltd.	Hydraulic Components	20.4 d	3.7 d	58.3%
Circuit & Cab Electronics	Electrical & Cabin	26 d	5.4 d	66%
Ironclad GET Works	Ground Engaging Tools	13.7 d	1.7 d	68.3%
Vantage Engine Systems	Engine & Powertrain	17.3 d	2.4 d	73.7%

Forecast

Seasonal + promo regression is the better planning basis

A trend + Fourier-seasonality + promotion regression averages 23.1% WMAPE per SKU x DC on a blind 8-week holdout, versus 24% for a 52-week moving average, and it removes the systematic under-forecast the moving average carries into promotion weeks. Every reviewed SKU shows a clear annual cycle and a measurable promotion lift.

Data quality

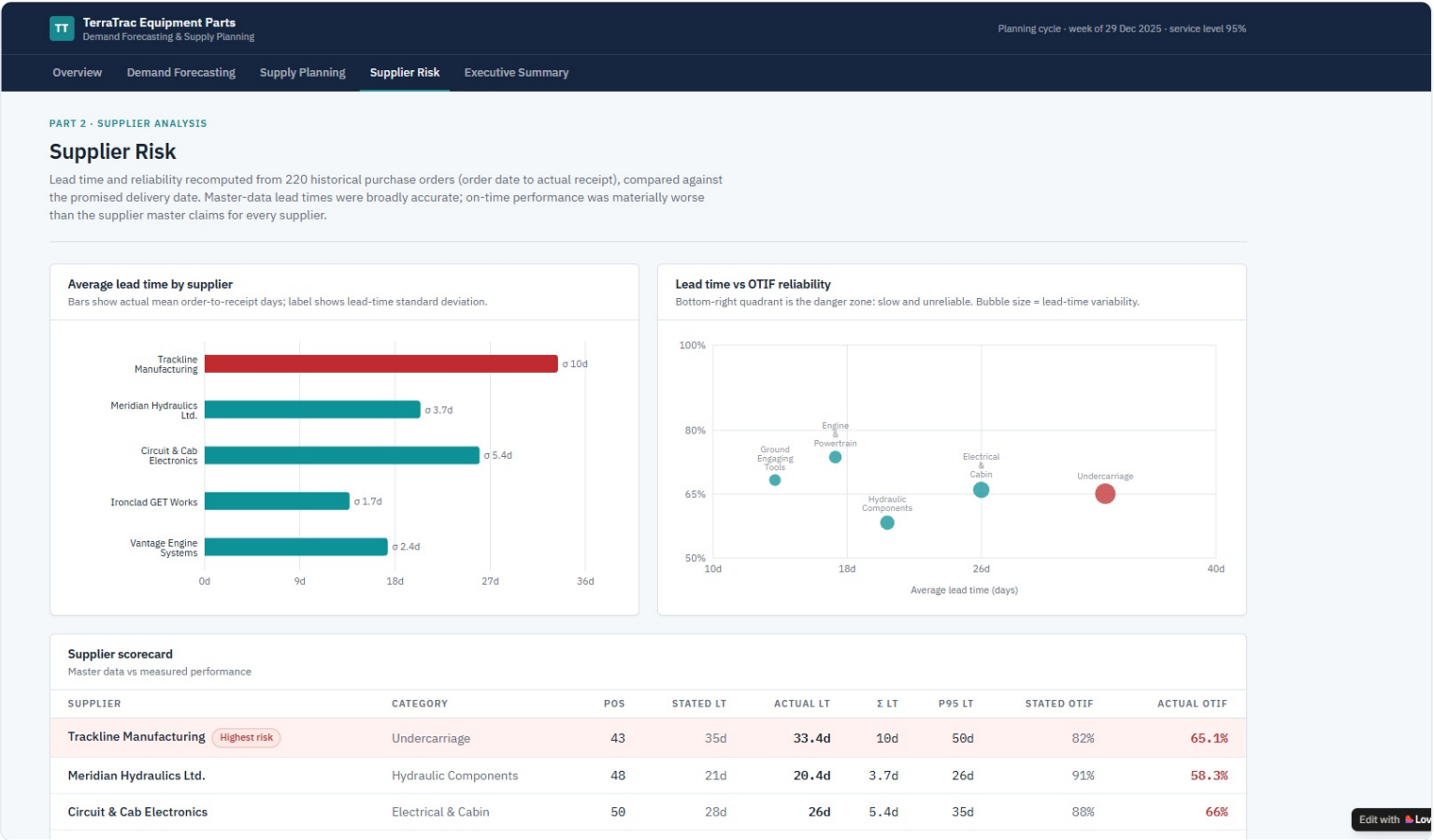
Undercarriage demand in weeks 41-46 is censored, not soft

The weeks 41-46 supply disruption at Chennai and Jakarta suppressed observed sell-through for all five Undercarriage SKUs by an average of 163%. Uncorrected, it drags the forecast down and understates safety stock for the least reliable supplier in the network.

Edit with Loveable

Overview tab — headline KPIs, network summary, and key findings, generated from the corrected dataset.

What it shows: forecast accuracy, positions below reorder point, recommended buy value, and worst supplier lead time, at a glance. **Why it matters:** a leadership audience needs the headline numbers before the detail. **Main insight:** 26 of 28 monitored positions sit below their reorder point — not a handful of isolated stockouts, a structural pattern. **Business implication:** replenishment discipline needs to change, not just this cycle's order quantities.



What the Analysis Found

INSIGHT 01

26 of 28 monitored SKU × warehouse positions are below their reorder point

16 of those cannot even cover a single replenishment lead time from on-hand stock alone. This isn't concentrated in one or two problem SKUs — it's network-wide, pointing to a missing reorder-point process rather than isolated bad luck.

INSIGHT 02

Trackline Manufacturing is a structural risk, not just a slow supplier

33.4-day average actual lead time with a 10-day standard deviation — more than 3× the variability of the most consistent supplier — and the sole source for the Undercarriage category. It also caused the demand-censoring event that had to be corrected in the forecast.

INSIGHT 03

Every supplier's real on-time rate is materially worse than its stated reliability figure

A gap of 12–33 percentage points across all five suppliers, verified against 220 actual purchase orders. This isn't a Trackline-specific issue — it suggests the stated reliability field shouldn't be trusted for planning without independent verification, for any supplier.

INSIGHT 04

A seasonality + promotion-aware forecast beats a naive baseline, honestly measured

23.1% WMAPE vs. 24.0% for a 52-week moving average, on an 8-week blind holdout. The margin is real but not dramatic — reported as such rather than overstated.

INSIGHT 05

Correcting censored demand changed the picture for the highest-risk category

The Undercarriage stockout-censoring correction raised measured demand during the affected weeks by 163%. Left uncorrected, both the forecast and the safety-stock buffer for the category with the least reliable supplier would have been understated — compounding the risk instead of catching it.

What TerraTrac Should Do With This

THIS WEEK

Release the catch-up replenishment plan

Supporting insight: 16 positions cannot cover one lead time from on-hand stock. **Recommendation:** place the \$8.16M replenishment plan across the 7 monitored SKUs now, respecting MOQ and pack size, and expedite the 16 critical positions ahead of the routine cycle. **Expected impact:** closes the most immediate stockout exposure before it becomes a customer-facing outage.

NEXT PLANNING CYCLE

Adopt the seasonality + promotion forecast as the planning basis

Supporting insight: it beats the current naive-average approach by a real, if modest, margin on a blind holdout. **Recommendation:** replace the 52-week moving average with this model, and track WMAPE and bias per SKU going forward rather than assuming it stays accurate. **Expected impact:** reduces both over- and under-forecasting, particularly around known promotional weeks.

30 DAYS

Re-parameterize safety stock for Undercarriage specifically

Supporting insight: Trackline's lead-time variability is the highest in the network by a wide margin. **Recommendation:** raise the service-level target for Undercarriage SKUs above the network-wide 95%, and size safety stock against a P95 lead time rather than the average. **Expected impact:** reduces stockout exposure in the single highest-risk category without over-buffering the rest of the portfolio.

90 DAYS

Qualify a secondary Undercarriage source

Supporting insight: Trackline is sole-sourced for the category and carries the network's highest lead-time risk. **Recommendation:** open a secondary-sourcing conversation, and use the verified on-time rate — not the supplier's own stated figure — in the next supplier review. **Expected impact:** reduces single-source exposure for the category most likely to cause a stockout.

What Was Actually Achieved

I scoped a supply-chain analytics problem, directed AI tools to build both the analysis pipeline (Python) and the product (an interactive dashboard via Lovable), and then did the part that's easy to skip: treated that output as a first draft to verify, not a finished answer to publish as-is.

That verification step is where the actual work happened. Cross-checking the supplier scorecard surfaced four suppliers reporting an identical, statistically implausible 70% on-time rate. Tracing it led to a one-line rounding-order bug. Checking the stockout-correction logic against the raw Stockout_Flag column found a second bug — a hardcoded date range one week off from what the data actually showed. Both were fixed, the analysis was re-run, the dashboard was rebuilt from the corrected output, and the fix was verified live on the published site before this document was written.

Every number in this document, and on the live dashboard, reflects the **corrected** analysis — not the first version the pipeline produced.

What the analysis revealed

A network running with no functional reorder-point discipline (26 of 28 positions below target), a single-sourced supplier carrying disproportionate lead-time risk, and a supplier reliability field that couldn't be trusted at face value for any vendor in the portfolio — not just the obviously slow one.

How this demonstrates analyst skill

Not by writing the most code by hand. By not accepting generated output at face value, verifying it against source data, finding where it was wrong, and being able to explain exactly why — then communicating the corrected findings clearly enough for a non-technical stakeholder to act on.

Final summary

A working, published data product with a documented, honest accuracy claim, a defensible inventory-planning methodology, and two concretely-identified and fixed analytical bugs — the kind of scrutiny that matters more in a real analytics job than any single chart.

What This Project Actually Shows

Only what's genuinely demonstrated by the work above — not a generic tools list.

Core analytical skills

Data quality assurance

Cross-verification against source data

Root-cause debugging

Inventory & supply-chain analytics

Safety stock / reorder-point modeling

Forecast accuracy evaluation (WMAPE, holdout validation)

Business-focused data storytelling

Tools & working methods

Python (pandas, NumPy)

Directing AI-assisted analytics pipelines

Directing AI application builders (Lovable)

Git / GitHub project management

Technical writing for non-technical audiences

What this project does *not* claim

Hand-written SQL · a manually-built dashboard frontend · Power BI or Excel-based visualization — none of these were part of this project, so none are claimed here.

PROJECT LINKS

Where to Find Everything

GITHUB PROFILE

<https://github.com/tahtabirrahman>

PROJECT REPOSITORY

<https://github.com/tahtabirrahman/terratrak-insights>

README

<https://github.com/tahtabirrahman/terratrak-insights/blob/main/README.md>

ANALYSIS SCRIPT

[scripts/supply_chain_analysis.py](#)

LIVE DASHBOARD

<https://trak-planner-insight.lovable.app>

JUPYTER NOTEBOOK

Not provided — analysis was run as a standalone script, not a notebook

TerraTrac Equipment Parts is a fictional case study. All company names, figures, and identifiers in the underlying dataset are synthetic and generated for this exercise; they do not represent any real organization, vendor, or business relationship.